Problems Sheet 5.

Problem 1. Construct the dual from the following problem
Maximize

$$z(x_1, x_2) = 2x_1 + 7x_2$$

subject to

$$\begin{cases} -2x_1 + 3x_2 \leq 14, & (1) \\ x_1 + x_2 \leq 8, & (2) \\ x_1 \geq 0, & (3) \\ x_2 \geq 0 & (4). \end{cases}$$

Using the graphical method find the primal and dual solutions.

Solution

First of all we need to construct the dual problem.
Primal problem is maximization problem. Consequently, in dual form we have minimization problem:
Minimize

$$w(y_1, y_2) = 14y_1 + 8y_2$$

.

In primal problem all variables are positive. Consequently,

$$\begin{cases} -2y_1 + y_2 \geq 2, & (1') \\ 3y_1 + y_2 \geq 7, & (2') \end{cases}$$

In (1), (2) we have $\leq$ sign. So

$$\begin{cases} y_1 \geq 0, & (3') \\ y_2 \geq 0 & (4'). \end{cases}$$

Therefore, the dual problem has the following form
Minimize

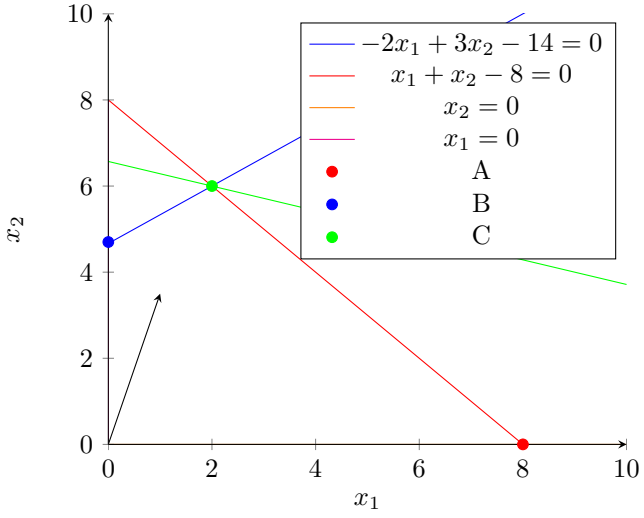
$$w(y_1, y_2) = 14y_1 + 8y_2$$

.

subject to

$$\begin{cases} -2y_1 + y_2 \geq 2, & (1') \\ 3y_1 + y_2 \geq 7, & (2') \\ y_1 \geq 0, & (3') \\ y_2 \geq 0 & (4'). \end{cases}$$

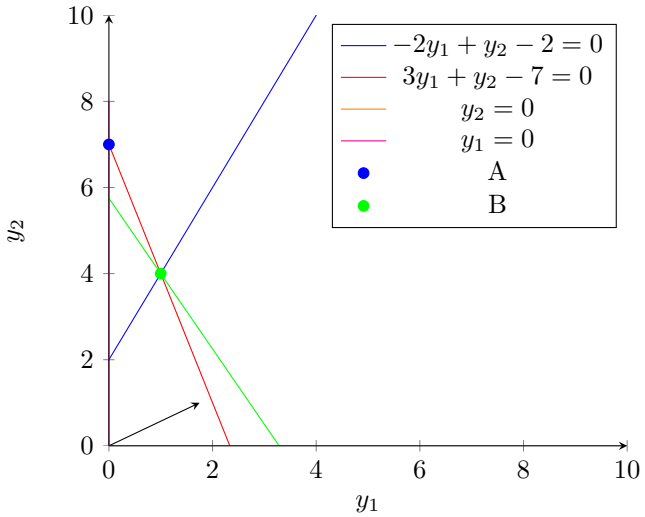
Using conditions (1)-(4) we get the following feasible Solution Space:



Consequently,

$$(x_1, x_2) = (2, 6), F(2, 6) = 46.$$

Using conditions $(1')$-$(4')$ we get the following feasible Solution Space:



Consequently,

$$(y_1, y_2) = (1, 4), w(1, 4) = 46.$$

Problem 2. Construct the dual from the following problem
Minimize

$$z(x_1, x_2) = -2x_1 - 3x_2$$

subject to

$$\begin{cases} -4x_1 + 2x_2 \geq 4, & (1) \\ x_1 + x_2 \geq 6, & (2) \\ x_1 \geq 0, & (3) \\ x_2 \geq 0 & (4). \end{cases}$$

Using the graphical method find the primal and dual solutions.

Solution

First of all we need to construct the dual problem.
Primal problem is minimization problem. Consequently, in the dual form we have maximization problem:
Maximize

$$w(y_1, y_2) = 4y_1 + 6y_2$$

.

In primal problem all variables are positive. Consequently,

$$\begin{cases} -4y_1 + y_2 \leq -2, & (1') \\ 2y_1 + y_2 \leq -3, & (2') \end{cases}$$

In (1), (2) we have $\geq$ sign. So

$$\begin{cases} y_1 \geq 0, & (3') \\ y_2 \geq 0 & (4'). \end{cases}$$

Therefore, the dual problem has the following form
Maximize

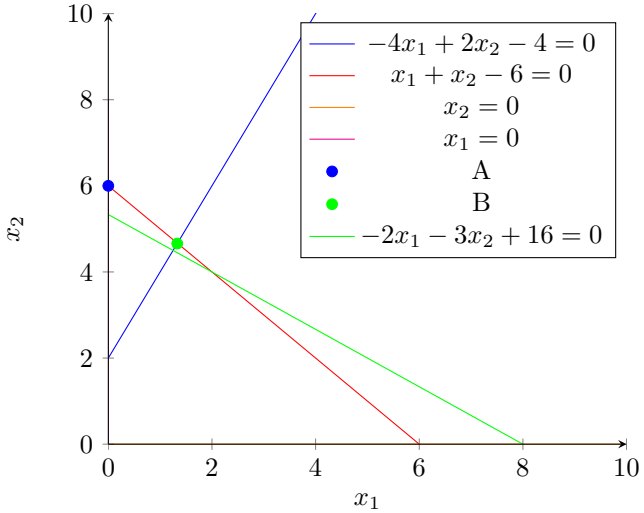
$$w(y_1, y_2) = 4y_1 + 6y_2$$

.

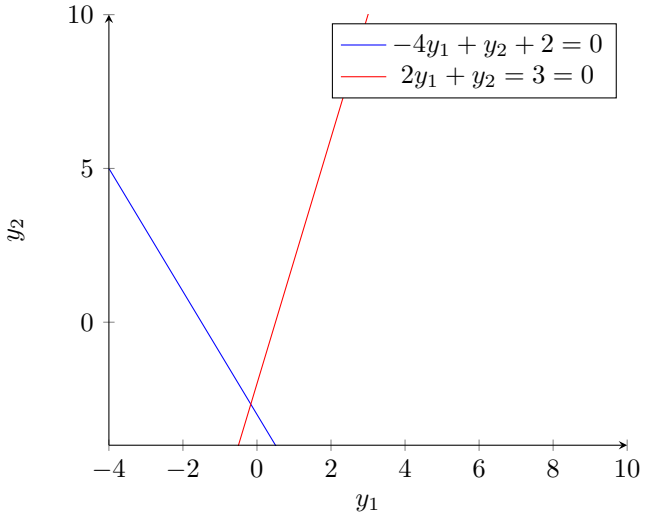
subject to

$$\begin{cases} -4y_1 + y_2 \leq -2, & (1') \\ 2y_1 + y_2 \leq -3, & (2') \\ y_1 \geq 0, & (3') \\ y_2 \geq 0 & (4'). \end{cases}$$

Using conditions (1)-(4) we get the following feasible Solution Space:



Consequently, our function is unbounded.
Using conditions (1')-(4') we get the following feasible Solution Space:



Consequently, the feasible set is empty.

Problem 3. Construct the dual from the following problem
Maximize

$$z(x_1, x_2, x_3) = x_1 + 2x_2 - x_3$$

subject to

$$\begin{cases} -x_1 + 4x_2 - 2x_3 \leq 12, & (1) \\ x_1 + x_2 + 2x_3 \leq 17, & (2) \\ 2x_1 - x_2 + 2x_3 = 4, & (3) \\ x_1 \geq 0, & (4) \\ x_2 \geq 0, & (5) \\ x_3 \geq 0. & (6) \end{cases}$$

Using the M-method find the primal and dual solutions.

Solution

First of all we need to construct the dual problem.
Primal problem is maximization problem. Consequently, in the dual form we have minimization problem:
minimize

$$w(y_1, y_2, y_3) = 12y_1 + 17y_2 + 4y_3$$

.

In primal problem all variables are positive. Consequently,

$$\begin{cases} -y_1 + y_2 + 2y_3 \geq 1, & (1') \\ 4y_1 + y_2 - y_3 \geq 2, & (2') \\ -2y_1 + 2y_2 + 2y_3 \geq -1, & (3') \end{cases}$$

In (1), (2) we have $\geq$ sign and in (3) we have $=$. So

$$\begin{cases} y_1 \geq 0, & (4') \\ y_2 \geq 0 & (5'). \end{cases}$$

Therefore, the dual problem has the following form
Minimize

$$w(y_1, y_2, y_3) = 12y_1 + 17y_2 + 4y_3$$

.

subject to

$$\begin{cases} -y_1 + y_2 + 2y_3 \geq 1, & (1') \\ 4y_1 + y_2 - y_3 \geq 2, & (2') \\ -2y_1 + 2y_2 + 2y_3 \geq -1, & (3') \\ y_1 \geq 0, & (4') \\ y_2 \geq 0 & (5'). \end{cases}$$

To find a solution of the dual problem we need to solve the primary problem. We will use M-method. The starting basic solution is $(x_1, x_2, x_3, s_1, s_2, R_1)$ = (0,0,0, 12,17,4).

Iteration	Basic	x_1	x_2	x_3	s_1	s_2	R_1	*Solution*	c_i
0	z	-1	-2	1	0	0	M		
	x_4	-1	4	-2	1	0	0	12	0
	x_5	1	1	2	0	1	0	17	0
	R_1	2	-1	2	0	0	1	4	-M

Before proceeding with the simplex method computations, the z-row must be made consistent with the rest of the tableau.

$$z_{new} = z_{old} + (0 \cdot s_1Row + 0 \cdot s_2Row - M \cdot R_1Row)z_{old}$$

where

$$z_{old} = (-1, -2, 1, 0, 0, M).$$

Iteration	Basic	x_1	x_2	x_3	s_1	s_2	R_1	*Solution*
0	z	**-201**	98	-199	0	0	0	-400
x_1 enters	s_1	-1	4	-2	1	0	0	12
	s_2	1	1	2	0	1	0	17
R_1 leaves	R_1	2	-1	2	0	0	1	4
1	z	0	**-5/2**	2	0	0	1/2	2
x_2 enters	s_1	0	7/2	-1	1	0	1/2	14
	s_2	0	3/2	1	0	1	-1/2	15
x_4 leaves	x_1	1	-1/2	1	0	0	1/2	2
2	z	0	0	9/7	5/7	0	6/7	12
	x_2	0	1	-2/7	2/7	0	1/7	4
	s_2	0	0	13/7	-3/7	1	-5/7	9
	x_1	1	0	6/7	1/6	0	4/7	4

Problem 4. Using "Advanced simplex method" solve the following problem
Maximize

$$z(x_1, x_2, x_3, x_4, x_5, x_6) = 9x_1 + 10x_2 + 16x_3$$

subject to

$$\begin{cases} 18x_1 + 15x_2 + 12x_3 + x_4 = 360, & (1) \\ 6x_1 + 4x_2 + 8x_3 + x_5 = 192, & (2) \\ 5x_1 + 3x_2 + 3x_3 + x_6 = 180, & (3) \\ x_1 \geq 0, & (4) \\ x_2 \geq 0 & (5) \\ x_3 \geq 0 & (6) \\ x_4 \geq 0 & (7) \\ x_5 \geq 0 & (8) \\ x_6 \geq 0 & (9) \end{cases}$$

Solution

We have

$$X_{B_0} = (x_4, x_5, x_6), \quad C_{B_0} = (0, 0, 0), \quad B_0 = (P_4, P_5, P_6), \quad B_0^{-1} = I.$$

Thus

$$X_{B_0} = B_0^{-1}b = (360, 192, 180)^T, z = C_{B_0}X_{B_0} = 0.$$

We need to find

$$\{z_j - c_j\}_{j=1,2,3} = C_{B_0}B_0^{-1}(P_1, P_2, P_3) - (c_1, c_2, c_3) = (-9, -10, -16).$$

Obviously,

$$\min\{-9, -10, -16\} = -16.$$

Thus, P_3 is the entering vector.
Since

$$B_0^{-1}P_3 = (12, 8, 3), \min\{\frac{360}{12}, \frac{192}{8}, \frac{180}{3}\} = \frac{192}{8},$$

P_5 becomes the leaving vector.

Iteration	Basic	A_1	A_2	A_3	P_3	$B_i^{-1}b$	C_{B_i}
x_3 enters	x_4	1	0	0	12	360	0
	x_5	0	1	0	8	192	0
x_5 leaves	x_6	0	0	1	3	180	0
	x_4	1	-3/2	0	0	72	0
	x_3	0	1/8	0	1	24	16
	x_6	0	-3/8	1	0	108	0

Iteration 1.
We have $C_{B_1} = (0, 16, 0)$,

$$B_1 = \begin{bmatrix} 1 & 12 & 0 \\ 0 & 8 & 0 \\ 0 & 3 & 1 \end{bmatrix},$$

$$B_1^{-1} = \begin{bmatrix} 1 & -3/2 & 0 \\ 0 & 1/8 & 0 \\ 0 & -3/8 & 1 \end{bmatrix}.$$

Note that we can use the table above to find inverse matrix or calculate it using well known methods.
Thus,

$$X_{B_1} = B_1^{-1}b = \begin{bmatrix} 1 & -3/2 & 0 \\ 0 & 1/8 & 0 \\ 0 & -3/8 & 1 \end{bmatrix} (360, 192, 180)^T = (72, 24, 108)^T,$$

$$C_{B_1}B_1^{-1} = (0, 16, 0) \begin{bmatrix} 1 & -3/2 & 0 \\ 0 & 1/8 & 0 \\ 0 & -3/8 & 1 \end{bmatrix} = (0, 2, 0),$$

$$\{z_j - c_j\}_{j=1,2,5} = C_{B_1}B_1^{-1}(P_1, P_2, P_5) - (c_1, c_2, c_5) =$$

$$= (0, 2, 0)\begin{bmatrix} 18 & 15 & 0 \\ 6 & 4 & 1 \\ 5 & 3 & 0 \end{bmatrix} - (9, 10, 0) =$$

$$= (0 \cdot 18 + 2 \cdot 6 + 0 \cdot 5 - 9, 0 \cdot 15 + 2 \cdot 4 + 0 \cdot 3 - 10, 0 \cdot 0 + 2 \cdot 1 + 0 \cdot 0 - 0) = (3, -2, 2).$$

Thus P_2 is the entering vector.

We have

$$B_1^{-1}P_2 = \begin{bmatrix} 1 & -3/2 & 0 \\ 0 & 1/8 & 0 \\ 0 & -3/8 & 1 \end{bmatrix}(15, 4, 3)^T = (9, 1/2, 3/2).$$

Obviously,

$$\min\{\frac{72}{9}, \frac{24}{1/2}, \frac{108}{3/2}\} = \frac{72}{9}.$$

and P_4 becomes the leaving vector.

Iteration	Basic	A_1	A_2	A_3	P_3	$B_i^{-1}b$	C_{B_1}
0	x_4	1	0	0	12	360	0
x_3 enters	x_5	0	1	0	8	192	0
x_5 leaves	x_6	0	0	1	3	180	0
1	x_4	1	-3/2	0	0	72	0
x_2 enters	x_3	0	1/8	0	1	24	16
x_4 leaves	x_6	0	-3/8	1	0	108	0
	x_2	1/9	-1/6	0		8	10
	x_3	-1/18	5/24	0		20	16
	x_6	-1/6	-1/8	1		96	0

Iteration 2.

Consequently, we have $C_{B_2} = (10, 16, 0)$

$$B_2 = \begin{bmatrix} 15 & 12 & 0 \\ 4 & 8 & 0 \\ 3 & 3 & 1 \end{bmatrix},$$

$$B_2^{-1} = \begin{bmatrix} 1/9 & -1/6 & 0 \\ -1/18 & 5/24 & 0 \\ -1/6 & -1/8 & 1 \end{bmatrix}.$$

Thus,

$$X_{B_2} = B_2^{-1}b = \begin{bmatrix} 1/9 & -1/6 & 0 \\ -1/18 & 5/24 & 0 \\ -3/2 & -1/8 & 1 \end{bmatrix}(360, 192, 180)^T = (8, 20, 96)$$

$$z = C_{B_2}X_{B_2} = 400,$$

$$C_{B_2}B_2^{-1} = (10, 16, 0)\begin{bmatrix} 1/9 & -1/6 & 0 \\ -1/18 & 5/24 & 0 \\ -3/2 & -1/8 & 1 \end{bmatrix} = (2/9, 5/3, 0),$$

$$\{z_j - c_j\}_{j=1,4,5} = C_{B_2}B_2^{-1}(P_1, P_4, P_5) - (c_1, c_4, c_5) =$$

$$(2/9, 5/3, 0)\begin{bmatrix} 18 & 1 & 0 \\ 6 & 0 & 1 \\ 5 & 0 & 0 \end{bmatrix} - (9, 0, 0)$$

$$= (\frac{2}{9} \cdot 18 + \frac{5}{3} \cdot 6 + 5 \cdot 0 - 9, \frac{2}{9} - 16, \frac{5}{3} \cdot 0 + 2 \cdot 1 + 0 \cdot 0 - 0) = (5, 2/9, 5/3).$$

Therefore,

$$z_{\max} = 400, (x_1, x_2, x_3, x_4, x_5, x_6) = (0, 8, 20, 0, 0, 96).$$

Problem 5. Using "Advanced simplex method" solve the following problem
Maximize

$$z(x_1, x_2, x_3, x_4, x_5, x_6) = 8x_2 + 7x_4 + x_6$$

subject to

$$\begin{cases} x_1 - 2x_2 - 3x_4 - 2x_6 = 12, & (1) \\ 4x_2 + x_3 - 4x_4 - 3x_6 = 12, & (2) \\ 5x_2 + 5x_4 + x_5 + x_6 = 25, & (3) \\ x_1 \geq 0, & (4) \\ x_2 \geq 0 & (5) \\ x_3 \geq 0 & (6) \\ x_4 \geq 0 & (7) \\ x_5 \geq 0 & (8) \\ x_6 \geq 0 & (9) \end{cases}$$

Solution

We have

$$X_{B_0} = (x_1, x_3, x_5), \quad C_{B_0} = (0, 0, 0), \quad B_0 = (P_1, P_3, P_5), \quad B_0^{-1} = I.$$

Thus

$$X_{B_0} = B_0^{-1} b = (12, 12, 25)^T, z = C_{B_0} X_{B_0} = 0.$$

We need to find

$$\{z_j - c_j\}_{j=2,4,6} = C_{B_0} B_0^{-1} (P_2, P_4, P_6) - (c_2, c_4, c_6) = (-8, -7, -1).$$

Obviously,

$$\min\{-8, -7, -1\} = -8.$$

Thus, P_2 is the entering vector.
Since

$$B_0^{-1} P_2 = (-2, 4, 5), \min\{\frac{12}{4}, \frac{25}{5}\} = \frac{12}{4},$$

P_3 becomes the leaving vector.

Iteration 1.
We have $X_{B_1} = (x_1, x_2, x_5)$, $C_{B_1} = (0, 8, 0)$,

$$B_1 = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 4 & 0 \\ 0 & 5 & 1 \end{bmatrix},$$

$$B_1^{-1} = \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & 1/4 & 0 \\ 0 & -5/4 & 1 \end{bmatrix}.$$

Note that we can use the table above to find inverse matrix or calculate it using well known methods.
Thus,

$$X_{B_1} = B_1^{-1} b = \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & 1/4 & 0 \\ 0 & -5/4 & 1 \end{bmatrix} (12, 12, 25)^T = (18, 3, 10),$$

$$C_{B_1} B_1^{-1} = (0, 8, 0) \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & 1/4 & 0 \\ 0 & -5/4 & 1 \end{bmatrix} = (0, 2, 0),$$

$$\{z_j - c_j\}_{j=3,4,6} = C_{B_1} B_1^{-1} (P_3, P_4, P_6) - (c_3, c_4, c_6) =$$

$$= (0, 2, 0) \begin{bmatrix} 0 & -3 & -2 \\ 1 & -4 & -3 \\ 0 & 5 & 1 \end{bmatrix} - (0, 7, 1) = (2, -15, -7).$$

Thus P_4 is the entering vector.

We have

$$B_1^{-1}P_4 = \begin{bmatrix} 1 & 1/2 & 0 \\ 0 & 1/4 & 0 \\ 0 & -5/4 & 1 \end{bmatrix} (-3,-4,5)^T = (-5,-1,10).$$

Obviously, P_5 becomes the leaving vector.

Iteration 2.

We have $X_{B_2} = (x_1, x_2, x_4)$, $C_{B_2} = (0,8,7)$,

$$B_2 = \begin{bmatrix} 1 & -2 & -3 \\ 0 & 4 & -4 \\ 0 & 5 & 5 \end{bmatrix},$$

$$B_2^{-1} = \begin{bmatrix} 1 & -1/8 & 1/2 \\ 0 & 1/8 & 1/10 \\ 0 & -1/8 & 1/10 \end{bmatrix}.$$

Thus,

$$X_{B_2} = B_2^{-1}b = \begin{bmatrix} 1 & -1/8 & 1/2 \\ 0 & 1/8 & 1/10 \\ 0 & -1/8 & 1/10 \end{bmatrix} (12,12,180)^T = (23,4,1)$$

$$z = C_{B_2}X_{B_2} = 39,$$

$$C_{B_2}B_2^{-1} = (0,8,7) \begin{bmatrix} 1 & -1/8 & 1/2 \\ 0 & 1/8 & 1/10 \\ 0 & -1/8 & 1/10 \end{bmatrix} = (0,1,4/5),$$

$$\{z_j - c_j\}_{j=3,5,6} = C_{B_2}B_2^{-1}(P_3,P_5,P_6) - (c_3,c_5,c_6) =$$

$$(2/9,5/3,0) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - (0,0,0) = (2/9,5/3,0).$$

Therefore, $z_{\max} = 39$ and $(x_1,x_2,x_3,x_4,x_5,x_6) = (23,4,0,1,0,0)$.